INCH-POUND

MIL-B-52301D
14 October 1992
SUPERSEDING
MIL-B-52301C
23 January 1984

MILITARY SPECIFICATION

BRACKET ASSEMBLY, CHAIN HOIST SUPPORT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bracket assembly and its components to be used with a trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of solicitation.

DRAWINGS

ME

TA13207E9734	- Bracket Assembly, Chain Hoist Support.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.1	- Carbon Steel Electrodes for Shielded Metal Arc Welding.
A 5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
D 1.1	- Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bracket assembly or components shall be as shown on top assembly TA13207E9734 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the bracket assembly. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The bracket assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the bracket assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the item may be newly fabricated from recovered materials to the maximum

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extent practicable, provided the item produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the item.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A 5.1 or AWS A 5.18, classification and size as required.

3.5 Identification marking. The bracket assembly or components, as applicable, shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The bracket assembly or components, as applicable, with the exception of the pin assembly and the bridge pin retainer assembly, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F. Exterior topcoat shall be camouflage green 383.

3.7 Stenciling. The nomenclature, National stock number, weight, and cube shall be stenciled on each bracket assembly. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 1/2 inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. Holes shall be drilled to full size or drilled undersize and reamed to full size through steel templates. All burrs shall be removed.

3.8.2 Welders and welding.

3.8.2.1 Welders or welding operator. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of ASME or AWS Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work

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shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.6).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The first article bracket assembly or components, as applicable, shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The bracket assembly or components, as applicable, shall be examined as specified herein for the following defects:

Major

101. Any part of component missing or not as specified (see 4.1.2)..
102. Material not as specified (see 3.3).
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1)..
105. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.1.2).
106. Welding electrodes as specified (see 3.4).
107. Workmanship not as specified (see 3.8.1).
108. Welders and welding operators not certified as specified (see 3.8.2.1)
109. Welding and welds not as specified (see 3.8.2).

Minor

201. Dimensions other than those identified by QAP data not as specified (see 3.1.1).
202. Identification marking incorrect, missing, or illegible (see 3.5).
203. Treatment and painting not as specified (see 3.6).
204. Stenciling not as specified (see 3.7).

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Major

111. Containers not as specified for level A or B. Each incorrect container shall constitute one defect (see 5.1.1 or 5.1.2 as applicable).
112. Components not secured within the boxes to prevent free movement as specified for level A or B (see 5.1.1 or 5.1.2 as applicable).
113. Strapping not as specified for level A (see 5.1.1).
114. Gross weight of wood boxes exceeds 250 pounds for level A or B (see 5.1.1 or 5.1.2 as applicable).
115. Gross weight or size of fiberboard boxes exceeds the box limitation for level B (see 5.1.2).
116. Fiberboard box not sealed and reinforced as specified for level B (see 5.1.2).
117. Packing not as specified for level C (see 5.1.3).
118. Marking missing, illegible, incorrect or incomplete for level A, B, or C (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The complete bracket assemblies shall be packed in close fitting boxes conforming to PPP-B-621, class 2, style optional; or boxes conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 250 pounds. Blocking and bracing to prevent movement and damage shall be provided in accordance with MIL-STD-1186. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

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5.1.1.1 Pin and bridge pin retainer assemblies. When procured separately from the bracket assembly, the pin assemblies or bridge pin retainer assemblies shall be packed as specified in 5.1.1 for complete bracket assemblies.

5.1.2 Level B. The complete bracket assemblies shall be packed as specified for level A, except the boxes shall be domestic type or class as applicable. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be as specified in 5.1.1.

5.1.2.1 Pin and bridge pin retainer assemblies. Assemblies of like description, procured separately from the bracket assembly, shall be packed together in close-fitting boxes conforming to PPP-B-621, class 1, style optional; PPP-B-601, domestic type, style optional; or PPP-B-636, V3c, style optional. The contents shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The gross weight or size of fiberboard boxes shall not exceed the limitations of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification. The gross weight of wood boxes shall not exceed 250 pounds and the box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Level C. The components comprising a complete bracket assembly, and the pin assembly and bridge pin retainer assembly, procured separately from the bracket assembly, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking. In addition to any special marking specified in the contract (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bracket assembly and its components are intended for use in floating and fixed bridges to support the chain hoist when the chain hoist is used to adjust the height of the transom in the trestle assembly.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).

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d. Level of packing required (see 5.1).
e. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.8.2.1	DI-MISC-80876	Welding Procedure Qualification Test Report.
3.8.2.1	DI-MISC-80875	Welding Procedures.

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6.6 Definition.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

Bracket assembly
Floating and fixed bridges
Trestle assembly

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Navy - MC
DLA - CS

Project 5420-0202

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-52301D	2. DOCUMENT DATE (YYMMDD) 921014

3. DOCUMENT TITLE
Bracket Assembly, Chain Hoist Support

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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